Liyan Gong

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Academic website: lg4147-ops.github.io

Education

New York University, Courant Institute of Mathematical Sciences *2025–Present*
M.S. in Mathematics

- **Passed Courant's Written Comprehensive Examinations**, May 2026 (the same examinations used for Ph.D. qualification). Subjects: Advanced Calculus, Complex Variables, and Linear Algebra.

Sun Yat-sen University *2021–2025*
B.S. in Mathematics and Applied Mathematics

Research Experience

Entrainment and fluctuations in circadian oscillators *2025–Present*
Courant Institute, New York University | Advisor: Charles S. Peskin

- Derived and verified an exact factorization linking protein–light correlation to waveform shape and the coherent variance fraction.
- Decomposed residual protein variance into along-cycle, transverse, and covariance contributions.
- Benchmarked chemical Langevin, hybrid, and Poisson tau-leap methods against Gillespie simulations.
- Implemented periodic linear-noise approximation (LNA) and reaction-resolved covariance analysis; constructed and tested a two-dimensional Floquet–LNA map.

Manuscript in preparation with Charles S. Peskin.

PINNs for the semiclassical Schrödinger equation *Undergraduate thesis*
Sun Yat-sen University | Advisor: Lihui Chai

- Implemented physics-informed neural networks (PINNs) in PyTorch to approximate WKB phase and amplitude functions through sequential training.
- Reconstructed semiclassical wavefunctions and compared them with Fourier spectral solutions; investigated the effects of sampling strategies and training settings on PINN training loss.

Multi-character text-to-video generation *Summer–Fall 2025*
Shanghai Jiao Tong University and Chongqing Institute of Artificial Intelligence

- Contributed to adapting the RPG framework to WAN, including attention-module debugging; investigated LoRA integration under the supervision of Prof. Peng Qi and Dr. Zhuang.
- Surveyed methods for preserving character identity and temporal consistency in multi-character video generation.

Talks and Presentations

Coherent response and residual fluctuations in a light-modulated circadian oscillator
AML Group Meeting, Courant Institute, New York University Oct. 16, 2026

Teaching Experience

New York University Instructor: Charles S. Peskin

- **Grader**, MATH-UA 144, Introduction to Computer Simulation Fall 2026
- **Grader**, MATH-UA 144, Introduction to Computer Simulation Spring 2026
- **Grader**, MATH-UA 268, Honors Ordinary Differential Equations Fall 2025

Selected Coursework

Graduate: Methods of Applied Mathematics; Fluid Dynamics; Partial Differential Equations; Atmospheric Dynamics; Information Theory for Statistics and Learning; Convex and Non-Smooth Optimization; Mechanics; Computational Electromagnetics; Geophysical Fluid Dynamics.
Undergraduate: Real Analysis; Complex Analysis; Functional Analysis I; Ordinary Differential Equations; Partial Differential Equations; Numerical Analysis; Stochastic Processes; Digital Image Processing.

Selected Honors

- Academic Exploration Award, Summer Research Program, CUHK-Shenzhen 2023
- Gold Medal, iGEM Competition, SYSU Software team 2022

- University Scholarship, Sun Yat-sen University — Two academic years

Technical skills and Languages

Technical skills: MATLAB, LaTeX, Python, C++, SQL, Microsoft Office.
Languages: Mandarin Chinese (native); English (proficient).